

Exploring the Relationship Between Heart Disease and its Risk Factors

Ethan Owusu-Bour
Computer Science Department
California Polytechnic University
Pomona, United States
eyowusubour@cpp.edu

Abstract—This project uses machine learning to explore the relationship between various risk factors related to heart disease using a synthetic dataset. Multiple biomarker diagnostics and lifestyle choices were included in the data and modeled using a random forest and a logistic regression model. Despite extensive cleaning and feature engineering, the results indicate that the dataset lacks the necessary structure to return meaningful predictions, severely limiting model performance. Further insights into model behavior and data structure are discussed.

Keywords—*biomarker diagnostics, random forest, logistic regression*

I. INTRODUCTION

The purpose of this project is to analyze relationships between various health indicators and risk factors that are commonly known to be associated with heart disease. The objective of the analysis include identifying which features contribute the most to predictive performance and exploring whether certain variables show exhibit stronger relationships with heart disease than expected.

1) Dataset Details

The data was obtained from Kaggle under the title “Heart Disease” containing 10,000 instances of data. The class distribution included 8,000 negative cases and 2,000 positive cases, reflecting a substantial class imbalance. Each instance included multiple demographic features, lifestyle factors, and biomarkers. Detailed descriptions of each variable are provided by the data source:

<https://www.kaggle.com/datasets/oktayrdeki/heart-disease/data>

- Age: The individual's age.
- Gender: The individual's gender (Male or Female).
- Blood Pressure: The individual's blood pressure (systolic).
- Cholesterol Level: The individual's total cholesterol level.
- Exercise Habits: The individual's exercise habits (Low, Medium, High).
- Smoking: Whether the individual smokes or not (Yes or No).

- Family Heart Disease: Whether there is a family history of heart disease (Yes or No).
- Diabetes: Whether the individual has diabetes (Yes or No).
- BMI: The individual's body mass index.
- High Blood Pressure: Whether the individual has high blood pressure (Yes or No).
- Low HDL Cholesterol: Whether the individual has low HDL cholesterol (Yes or No).
- High LDL Cholesterol: Whether the individual has high LDL cholesterol (Yes or No).
- Alcohol Consumption: The individual's alcohol consumption level (None, Low, Medium, High).
- Stress Level: The individual's stress level (Low, Medium, High).
- Sleep Hours: The number of hours the individual sleeps.
- Sugar Consumption: The individual's sugar consumption level (Low, Medium, High).
- Triglyceride Level: The individual's triglyceride level.
- Fasting Blood Sugar: The individual's fasting blood sugar level.
- CRP Level: The C-reactive protein level (a marker of inflammation).
- Homocysteine Level: The individual's homocysteine level (an amino acid that affects blood vessel health).
- Heart Disease Status: The individual's heart disease status (Yes or No).

II. METHODOLOGY

The following explains the preprocessing and machine learning steps in the pipeline.

A. Preprocessing

To begin, I looked at missing values. While every column contained missing values, there weren't many overall, except for

‘Alcohol Consumption’ which had over 2,500 missing values. Considering the dataset only contained 10,000 values, this was concerning. I initially considered removing the feature all together but I decided to look back at the data source since I believed there may have been a value for no alcohol consumption. My suspicions were confirmed and it would seem the dataset just has ‘Nan’ rather than an explicit ‘None’. I replaced those with their correct value then removed the rest of the missing values resulting in a 5% loss of data which I deemed acceptable.

Each feature, aside from ‘Alcohol Consumption’, contained about 25 missing values, which in hindsight, is unrealistic. The same goes for the engineered class imbalance; however, that is something that can genuinely happen and so I took the liberty of assuming the data was structured that way to help practice how to deal with it and I dealt with it in two ways. The first way preserved the structure of the distribution when performing the test train split, and the second way took the positive responses and formed a new data frame around their counts by randomly choosing negative instances to form an equal distribution. Both ways were used in the models.

The last thing done, after splitting to avoid data leak, was encoding the data. The data contained numerical, nominal, and ordinal features. Each type was separated into a list and processed. The numerical features were scaled to mean zero and standard deviation of one. This is convenient to do since I planned on running regression which assumes linearity of the data and has assumptions on the error terms. The nominal data was all binary and thus was encoded as zeros and ones. The ordinal data had an obvious structure and so they were encoded as 0, 1, 2, and 3 in the case of ‘Alcohol Consumption’. The distance assumed on the data is acceptable because all values were described as none, low, medium, or high, which follows an order but doesn’t imply any complicated distance. After transforming the data, the models were ready to be used.

B. Modeling

The following explains my approach to the chosen machine learning models.

1) Random Forest

Since there were 20 features, I had decided to use a decision trees to prune the data for me. I would do this by building a random forest and looking at the information gain of each predictor. Only in this case, the trees use Gini instead of entropy. This means that importance will be returned as a diagnostic rather than information gain. That being said, trees tend to overvalue continuous features, and so I refrained from over interpreting the results and compared with the full regression model before pruning.

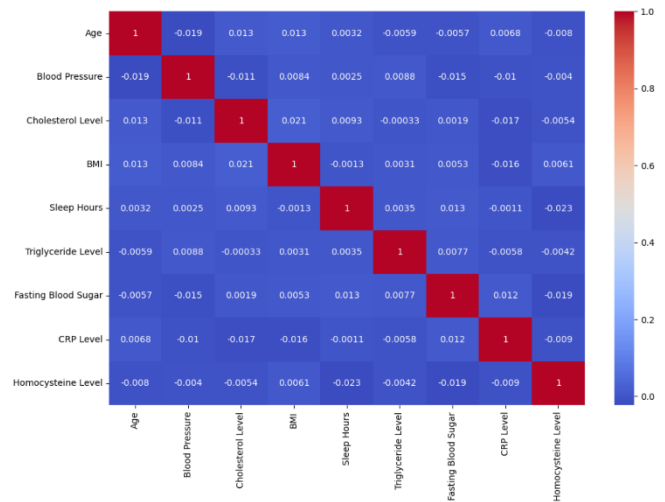
2) Logistic Regression

To perform statistical inference, logistic regression was implemented using statsmodels GLM framework, which provides detailed model summaries such as p-values and measures of error and variability. The model that included every predictor severely underperformed. Most coefficients were statistically insignificant when evaluated one the information they add to the model, and the model only explains 2% of the variation in the target variable. At

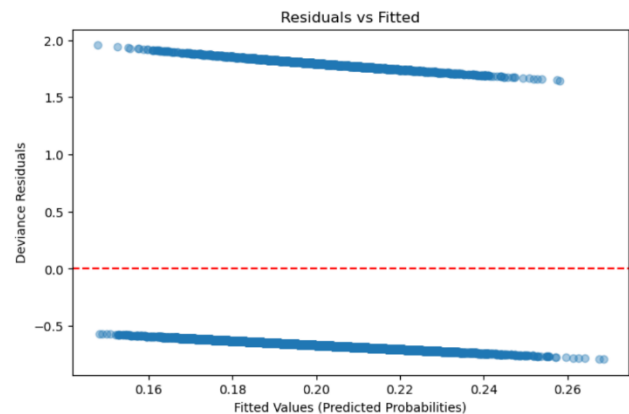
this point, I was confused because the result did not make any sense. However, I decided to continue testing and look into common reasons why a regression might fail.

III. RESULTS

One reason why a regression might fail is correlation between variables. I explored this by looking at the correlation heatmap. Much to my surprise, nothing was correlated. I also tested for variance inflation factors and got values close to one for every variable. This should be impossible considering the type of data we’re working with, especially with obvious correlations such as the numerical value for blood pressure and the categorical values saying whether an individual has high blood pressure. I let it be and went on to look at the residual plots.



The residual vs fitted plot revealed that the fitted probabilities lie within a tight range of about 0.15 to 0.26. This can be an indicator of weak predictors. Not much of a surprise consider the regression summary said the same thing, but this result still defies common knowledge.



Going forward, I decided to use the random forest to predict and return the accuracy. At first, I got an accuracy of 0.8; however, due to class imbalance, there were no positives being predicted. I reran the same steps of the analysis, but I used a training split with an equal representation of each class and got an accuracy of 50%. The model performs no better than taking

a random guess. After all this, everything pointed to the data being the issue rather than the models. Even if the data was nonlinear explaining why regression failed, random forest doesn't require and yet it still returned poor performance metrics. Returning to the data led me to discover that other people had similar discrepancies with the data and concluded that it was synthetic.

IV. CONCLUSION

In the end, I was not able to find any meaningful relationships between the features and the response, but that was due to the data rather than the models. It was good that the analysis was run on a topic as common as factors relating to heart disease otherwise I may have believed there was no connection. It's unfortunate that the data was only recently discovered to be unreliable and poorly constructed; however, that doesn't take away from the overall analysis. To be able to compare my findings to other reliable sources or simply ground

them on common knowledge provides a baseline for what results are to be expected. Rather than discrediting the model, it's prudent to know how the data was sourced and where it was obtained because that is where the fault truly lies. In my case, that was only possible to know after running everything. Aside from issues within the data, I was able to gain experience working with class imbalances and testing two different ways to deal with it. Preserving the distribution may be realistic to how the data is found and may provide deeper insights. Something like naïve bayes may benefit from this since that model has to do with probabilities. The other way, sampling an equal distribution, makes the data easier to work with and minimizes any issues the model might run into, but there is also a loss of information. It was also a good thing to look at graphs rooted in analysis rather than pure visualization since they provide more information and ultimately led me to check back on the data and question it. Overall, despite the data being unrealistic, it was a good experience and now I know to watch out for fake data and possibly how to detect it.